# Asterix category 021 - ADS-B Target Reports

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## Preamble

Surveillance data exchange. ADS-B Target Reports.

## Description of standard data items

### I021/010 - Data Source Identification

definition: Identification of the ADS-B station providing information.

Group

#### I021/010/SAC - System Area Code

Element
bit size: 8
Raw Content

#### I021/010/SIC - System Identification Code

Element
bit size: 8
Raw Content

**Note:**

- The up-to-date list of SACs is published on the EUROCONTROL ASTERIX Web Site (http://www.eurocontrol.int/services/system-area-code-list).

### I021/020 - Emitter Category

definition: Characteristics of the originating ADS-B unit.

Element
bit size: 8
Values:

**1:** Light aircraft <= 7000 kg
**2:** Reserved
**3:** 7000 kg < Medium aircraft < 136000 kg
**4:** Reserved
**5:** 136000 kg <= Heavy aircraft
**6:** Highly manoeuvrable (5g acceleration capability) and high speed (>400 knots cruise)
**7:** Reserved
**8:** Reserved
**9:** Reserved
**10:** Rotocraft
**11:** Glider / sailplane
**12:** Lighter-than-air
**13:** Unmanned aerial vehicle
**14:** Space / transatmospheric vehicle
**15:** Ultralight / handglider / paraglider
**16:** Parachutist / skydiver

**17:** Reserved
**18:** Reserved
**19:** Reserved
**20:** Surface emergency vehicle
**21:** Surface service vehicle
**22:** Fixed ground or tethered obstruction
**23:** Reserved
**24:** Reserved

### I021/030 - Time of Day

definition: Time of applicability (measurement) of the reported position, in the form of elapsed time since last midnight, expressed as UTC.

Element
bit size: 24
Unsigned quantity
LSB = $1/2^7$ s ≈ $7.8125e-3$ s
unit: ”s”

The time of the day value is reset to zero at every midnight.

### I021/032 - Time of Day Accuracy

definition: The maximum difference between the actual time of applicability of the reported position and the time reported in the Time of Day item (I021/030).

Element
bit size: 8
Unsigned quantity
LSB = $1/2^8$ s ≈ $3.90625e-3$ s
unit: ”s”

### I021/040 - Target Report Descriptor

definition: Type and characteristics of the data as transmitted by a system.

Group

#### I021/040/DCR - Differential Correction

Element
bit size: 1
Values:
**0:** No differential correction (ADS-B)
**1:** Differential correction (ADS-B)

#### I021/040/GBS - Ground Bit Setting

Element
bit size: 1
Values:
**0:** Ground Bit not set
**1:** Ground Bit set

#### I021/040/SIM - Simulated Target

Element
bit size: 1
Values:
**0:** Actual target report
**1:** Simulated target report

**I021/040/TST - Test Target**

Element
bit size: 1
Values:

- **0:** Default
- **1:** Test Target

**I021/040/RAB - Report Type**

Element
bit size: 1
Values:

- **0:** Report from target transponder
- **1:** Report from field monitor (fixed transponder)

**I021/040/SAA - Selected Altitude Available**

Element
bit size: 1
Values:

- **0:** Equipment capable to provide Selected Altitude
- **1:** Equipment not capable to provide Selected Altitude

**I021/040/SPI - Special Position Identification**

Element
bit size: 1
Values:

- **0:** Absence of SPI
- **1:** Special Position Identification

Spare bits: 1

**I021/040/ATP - Address Type**

Element
bit size: 3
Values:

- **0:** Non unique address
- **1:** 24-Bit ICAO address
- **2:** Surface vehicle address
- **3:** Anonymous address
- **4:** Reserved for future use
- **5:** Reserved for future use
- **6:** Reserved for future use
- **7:** Reserved for future use

**I021/040/ARC - Altitude Reporting Capability**

Element
bit size: 2
Values:

- **0:** Unknown
- **1:** 25 ft
- **2:** 100 ft

Spare bits: 3

**I021/070 - Mode 3/A Code in Octal Representation**

definition: Mode-3/A code converted into octal representation.
Group

**I021/070/V**

Element
bit size: 1
Values:

**0:** Code validated
**1:** Code not validated

**I021/070/G**

Element
bit size: 1
Values:

**0:** Default
**1:** Garbled code

**I021/070/L**

Element
bit size: 1
Values:

**0:** Mode-3/A code derived during last update
**1:** Mode-3/A code not extracted during the last update

Spare bits: 1

**I021/070/MODE3A - Mode-3/A Reply in Octal Representation**

Element
bit size: 12
Octal string (3-bits per char)

Notes:

1. Bit 15 (G) is set to one when an error correction has been attempted.
2. Bit 16 (V) is normally set to zero, but can exceptionally be set to one to indicate a non-validated Mode-3/A code (e.g. alert condition detected, but new Mode-3/A code not successfully extracted)

## I021/080 - Target Address

definition: Target address (emitter identifier) assigned uniquely to each target.

Element
bit size: 24
Raw Content

## I021/090 - Figure of Merit

definition: ADS figure of merit (FOM) provided by the aircraft avionics.

Group

**I021/090/AC - ACAS Capabilities**

Element
bit size: 2
Values:

**0:** Unknown
**1:** ACAS not operational
**2:** ACAS operartional
**3:** Invalid

**I021/090/MN - Multiple Navigation Aids**

Element
bit size: 2
Values:

**0:** Unknown
**1:** Multiple Navigation not operational
**2:** Multiple Navigation operartional

**3:** Invalid

**I021/090/DC - Differential Correction**

Element
bit size: 2
Values:

**0:** Unknown
**1:** Differential Correction
**2:** NO Differential Correction
**3:** Invalid

Spare bits: 6

**I021/090/PA - Position Accuracy**

Element
bit size: 4
Signed quantity
LSB = 1 ≈ 1.0
unit: ""

**Note:** bits-4/1 (PA) code the "Navigational Uncertainty Categories – Position" as described in the ADS-B MASPS [Ref. 3]

**I021/095 - Velocity Accuracy**

definition: Velocity uncertainty category of the least accurate velocity

Element
bit size: 8
Raw Content

**Note:** bits-8/1 code the "Navigational Uncertainty Categories – Velocity" as described in the ADS-B MASPS [Ref. 3]

**I021/110 - Trajectory Intent**

definition: Reports indicating the 4D intended trajectory of the aircraft.

Compound

**I021/110/TIS - Trajectory Intent Status**

Extended

**I021/110/TIS/NAV**

Element
bit size: 1
Values:

**0:** Trajectory Intent Data is available for this aircraft
**1:** Trajectory Intent Data is not available for this aircraft

**I021/110/TIS/NVB**

Element
bit size: 1
Values:

**0:** Trajectory Intent Data is valid
**1:** Trajectory Intent Data is not valid

Spare bits: 5
*(FX) - extension bit*

**I021/110/TID - Trajectory Intent Data**

Repetitive
Regular, 1 byte(s) REP field size.
Group

**I021/110/TID/TCA**

Element
bit size: 1
Values:

- **0:** TCP number available
- **1:** TCP number not available

**I021/110/TID/NC**

Element
bit size: 1
Values:

- **0:** TCP compliance
- **1:** TCP non-compliance

**I021/110/TID/TCPN**

description: Trajectory Change Point number

Element
bit size: 6
Raw Content

**I021/110/TID/ALT - Altitude in Two's Complement Form**

Element
bit size: 16
Signed quantity
LSB = 10 ft ≈ 10.0 ft
unit: ”ft”
>= $-1500.0$
<= $150000.0$

**I021/110/TID/LAT - In WGS.84 in Two's Complement**

Element
bit size: 24
Signed quantity
LSB = $180/2^23$ ° ≈ $2.1457672119140625e-5$ °
unit: ”°”
>= $-90.0$
<= $90.0$

**I021/110/TID/LON - In WGS.84 in Two's Complement**

Element
bit size: 24
Signed quantity
LSB = $180/2^23$ ° ≈ $2.1457672119140625e-5$ °
unit: ”°”
>= $-180.0$
< $180.0$

**I021/110/TID/PT - Point Type**

Element
bit size: 4
Values:

- **0:** Unknown
- **1:** Fly by waypoint (LT)
- **2:** Fly over waypoint (LT)
- **3:** Hold pattern (LT)
- **4:** Procedure hold (LT)
- **5:** Procedure turn (LT)
- **6:** RF leg (LT)
- **7:** Top of climb (VT)
- **8:** Top of descent (VT)

**9:** Start of level (VT)
**10:** Cross-over altitude (VT)
**11:** Transition altitude (VT)

**I021/110/TID/TD**

Element
bit size: 2
Values:

**0:** N/A
**1:** Turn right
**2:** Turn left
**3:** No turn

**I021/110/TID/TRA**

description: Turn Radius Availability

Element
bit size: 1
Values:

**0:** TTR not available
**1:** TTR available

**I021/110/TID/TOA**

Element
bit size: 1
Values:

**0:** TOV available
**1:** TOV not available

**I021/110/TID/TOV - Time Over Point**

Element
bit size: 24
Unsigned quantity
LSB = 1 s ≈ 1.0 s
unit: ”s”

**I021/110/TID/TTR - TCP Turn Radius**

Element
bit size: 16
Unsigned quantity
LSB = $1/100$ NM ≈ $1.0e-2$ NM
unit: ”NM”
>= 0.0
<= 655.35

Notes:

1. NC is set to one when the aircraft will not fly the path described by the TCP data.
2. TCP numbers start from zero.
3. LT = Lateral Type
4. VT = Vertical Type
5. TOV gives the estimated time before reaching the point. It is defined as the absolute time from midnight.
6. TOV is meaningful only if TOA is set to 1.

**I021/130 - Position in WGS-84 Co-ordinates**

definition: Calculated Position in WGS-84 Co-ordinates with a resolution of 180/(2^25) degrees.

Group

**I021/130/LAT - Latitude**

Element
bit size: 32
Signed quantity
LSB = $180/2^25$ ° ≈ $5.36441802978515625e-6$ °
unit: ”°”
>= $-90.0$
<= $90.0$

**I021/130/LON - Longitude**

Element
bit size: 32
Signed quantity
LSB = $180/2^25$ ° ≈ $5.36441802978515625e-6$ °
unit: ”°”
>= $-180.0$
< $180.0$

Notes:

1. Positive longitude indicates East. Positive latitude indicates North.
2. The LSB provides a resolution at least better than 0.6m.

### I021/131 - Signal Amplitude

definition: Relative strength of received signal.

Element
bit size: 8
Raw Content

**Note:** The value is implementation-dependent, 0 being the minimum detectable level for that system.

### I021/140 - Geometric Altitude

definition: Vertical distance between the target and the projection of its position on the earth’s ellipsoid, as defined by WGS84, in two’s complement form.

Element
bit size: 16
Signed quantity
LSB = $25/2^2$ ft ≈ $6.25$ ft
unit: ”ft”
>= $-1500.0$
< $150000.0$

**Note:**

1. LSB is required to be less than 10 ft by ICAO.

### I021/145 - Flight Level

definition: Flight Level from barometric measurements,not QNH corrected, in two’s complement form.

Element
bit size: 16
Signed quantity
LSB = $1/2^2$ FL ≈ $0.25$ FL
unit: ”FL”
>= $-15.0$
< $1500.0$

### I021/146 - Intermediate State Selected Altitude

definition: The short-term vertical intent as described by either the FMS selected altitude, the Altitude Control Panel Selected Altitude, or the current aircraft altitude according to the aircraft's mode of flight.

Group

#### I021/146/SAS - Source Availability

Element
bit size: 1
Values:

- **0:** No source information provided
- **1:** Source Information provided

#### I021/146/SRC - Source

Element
bit size: 2
Values:

- **0:** Unknown
- **1:** Aircraft Altitude (Holding Altitude)
- **2:** MCP/FCU Selected Altitude
- **3:** FMS Selected Altitude

#### I021/146/ALT - Altitude

Element
bit size: 13
Signed quantity
LSB = $25$ ft $\approx 25.0$ ft
unit: ”ft”
>= $-1300.0$
< $100000.0$

### I021/148 - Final State Selected Altitude

definition: The vertical intent value that corresponds with the ATC cleared altitude, as derived from the Altitude Control Panel (MCP/FCU).

Group

#### I021/148/MV - Manage Vertical Mode

Element
bit size: 1
Values:

- **0:** Not active
- **1:** Active

#### I021/148/AH - Altitude Hold Mode

Element
bit size: 1
Values:

- **0:** Not active
- **1:** Active

#### I021/148/AM - Approach Mode

Element
bit size: 1
Values:

- **0:** Not active
- **1:** Active

#### I021/148/ALT - Altitude

Element
bit size: 13
Signed quantity
LSB = $25$ ft ≈ $25.0$ ft
unit: ”ft”
>= $-1300.0$
< $100000.0$

### I021/150 - Air Speed

definition: Calculated Air Speed (Element of Air Vector).

Group

#### I021/150/IM

Element
bit size: 1
Values:

**0:** Air Speed = IAS, LSB (Bit-1) = 2 -14 NM/s
**1:** Air Speed = Mach, LSB (Bit-1) = 0.001

#### I021/150/AS - Air Speed (IAS or Mach)

Element
bit size: 15
Depending on: (150/IM)

**(0):** Unsigned quantity
LSB = $1/2^14$ NM/s ≈ $6.103515625e-5$ NM/s
unit: ”NM/s”

**(1):** Unsigned quantity
LSB = $1/1000$ Mach ≈ $1.0e-3$ Mach
unit: ”Mach”

Default:

Raw Content

### I021/151 - True Airspeed

definition: True Air Speed.

Element
bit size: 16
Unsigned quantity
LSB = $1$ kt ≈ $1.0$ kt
unit: ”kt”

### I021/152 - Magnetic Heading

definition: Magnetic Heading (Element of Air Vector).

Element
bit size: 16
Unsigned quantity
LSB = $360/2^16$ ° ≈ $5.4931640625e-3$ °
unit: ”°”

### I021/155 - Barometric Vertical Rate

definition: Barometric Vertical Rate, in two’s complement form.

Element
bit size: 16
Signed quantity
LSB = $25/2^2$ ft/min ≈ $6.25$ ft/min
unit: ”ft/min”

### I021/157 - Geometric Vertical Rate

definition: Geometric Vertical Rate, in two’s complement form, with reference to WGS-84.

Element
bit size: 16
Signed quantity
LSB = $25/2^2$ ft/min ≈ $6.25$ ft/min
unit: ”ft/min”

### I021/160 - Ground Vector

definition: Ground Speed and Track Angle elements of Ground Vector.

Group

#### I021/160/GS - Ground Speed in Two's Complement Form Referenced to WGS84

Element
bit size: 16
Signed quantity
LSB = $1/2^14$ NM/s ≈ $6.103515625e-5$ NM/s
unit: ”NM/s”
>= $0.0$
< $2.0$

#### I021/160/TA - Track Angle

Element
bit size: 16
Unsigned quantity
LSB = $360/2^16$ ° ≈ $5.4931640625e-3$ °
unit: ”°”

### I021/165 - Rate Of Turn

definition: Rate of Turn, in two’s complement form.

Extended

#### I021/165/TI - Turn Indicator

Element
bit size: 2
Values:

- **0:** Not available
- **1:** Left
- **2:** Right
- **3:** Straight

Spare bits: 5
*(FX) - extension bit*

#### I021/165/ROT - Rate of Turn

Element
bit size: 7
Signed quantity
LSB = $1/2^2$ °/s ≈ 0.25 °/s
unit: "°/s"
<= 15.0

*(FX) - extension bit*

**Notes:**

1. A positive value represents a right turn, whereas a negative value represents a left turn.
2. Value 15 means 15 °/s or above.

### I021/170 - Target Identification

definition: Target (aircraft or vehicle) identification in 8 characters, as reported by the target.

Element
bit size: 48
ICAO string (6-bits per char)

### I021/200 - Target Status

definition: Status of the target

Element
bit size: 8
Values:

**0:** No emergency / not reported
**1:** General emergency
**2:** Lifeguard / medical
**3:** Minimum fuel
**4:** No communications
**5:** Unlawful interference

### I021/210 - Link Technology Indicator

definition: Indication of which ADS link technology has been used to send the target report.

Group

Spare bits: 3

#### I021/210/DTI - Cockpit Display of Traffic Information

Element
bit size: 1
Values:

**0:** Unknown
**1:** Aircraft equiped with CDTI

#### I021/210/MDS - Mode-S Extended Squitter

Element
bit size: 1
Values:

**0:** Not used
**1:** Used

#### I021/210/UAT - UAT

Element
bit size: 1
Values:

**0:** Not used
**1:** Used

#### I021/210/VDL - VDL Mode 4

Element
bit size: 1
Values:

**0:** Not used
**1:** Used

#### I021/210/OTR - Other Technology

Element
bit size: 1
Values:

**0:** Not used
**1:** Used

### I021/220 - Met Information

definition: Meteorological information.

Compound

#### I021/220/WS - Wind Speed

Element
bit size: 16
Unsigned quantity
LSB = 1 kt ≈ 1.0 kt
unit: ”kt”
>= 0.0
<= 300.0

#### I021/220/WD - Wind Direction

Element
bit size: 16
Unsigned quantity
LSB = 1 ° ≈ 1.0 °
unit: ”°”
>= 1.0
<= 360.0

#### I021/220/TMP - Temperature

Element
bit size: 16
Signed quantity
LSB = $1/2^2$ °C ≈ 0.25 °C
unit: ”°C”
>= −100.0
<= 100.0

#### I021/220/TRB - Turbulence

Element
bit size: 8
Unsigned integer
>= 0.0
<= 15.0

### I021/230 - Roll Angle

definition: The roll angle, in two's complement form, of an aircraft executing a turn.

Element
bit size: 16
Signed quantity
LSB = $1/100$ ° ≈ $1.0e-2$ °
unit: "°"
>= $-180.0$
<= $180.0$

**Notes:**

1. Negative Value indicates "Left Wing Down".
2. Resolution provided by the technology "1090 MHz Extended Squitter" is 1 degree.

#### I021/RE - Reserved Expansion Field

definition: Expansion

Explicit (ReservedExpansion)

#### I021/SP - Special Purpose Field

definition: Special Purpose Field

Explicit (SpecialPurpose)

## User Application Profile

- 1: `I021/010` - Data Source Identification
- 2: `I021/040` - Target Report Descriptor
- 3: `I021/030` - Time of Day
- 4: `I021/130` - Position in WGS-84 Co-ordinates
- 5: `I021/080` - Target Address
- 6: `I021/140` - Geometric Altitude
- 7: `I021/090` - Figure of Merit
- (FX) - Field extension indicator
- 8: `I021/210` - Link Technology Indicator
- 9: `I021/230` - Roll Angle
- 10: `I021/145` - Flight Level
- 11: `I021/150` - Air Speed
- 12: `I021/151` - True Airspeed
- 13: `I021/152` - Magnetic Heading
- 14: `I021/155` - Barometric Vertical Rate
- (FX) - Field extension indicator
- 15: `I021/157` - Geometric Vertical Rate
- 16: `I021/160` - Ground Vector
- 17: `I021/165` - Rate Of Turn
- 18: `I021/170` - Target Identification
- 19: `I021/095` - Velocity Accuracy
- 20: `I021/032` - Time of Day Accuracy
- 21: `I021/200` - Target Status
- (FX) - Field extension indicator
- 22: `I021/020` - Emitter Category
- 23: `I021/220` - Met Information
- 24: `I021/146` - Intermediate State Selected Altitude
- 25: `I021/148` - Final State Selected Altitude
- 26: `I021/110` - Trajectory Intent
- 27: `I021/070` - Mode 3/A Code in Octal Representation
- 28: `I021/131` - Signal Amplitude

- (FX) - Field extension indicator
- *Spare*
- *Spare*
- *Spare*
- *Spare*
- *Spare*
- 34: `I021/RE` - Reserved Expansion Field
- 35: `I021/SP` - Special Purpose Field
- (FX) - Field extension indicator